Underwater Time-of-Flight Camera for Remotely Operated Vehicle Applications

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Abstract: Grove City College researchers have worked on development of a time-of-flight (ToF) camera for 3D imaging aboard underwater vehicles. We have modified a commercial camera for use underwater with green illuminators and have packaged and deployed this camera as payload aboard a BlueROV2. We describe the system hardware, including the 525 nm laser diode illuminator modules, the signal breakout board, and the wide field of view optics. We also show imaging and ranging results from laboratory and field tests with the 3D camera and discuss the challenges introduced by absorption and scattering in turbid water. We show preliminary results from a method for improving imaging in turbid water via backscatter subtraction.

Background: Time-of-Flight Cameras for 3D Imaging Underwater

Time-of-flight (ToF) cameras simultaneously generate amplitude and depth images by measuring both the reflectivity and distance of every pixel of every frame (Figure 1 and Figure 2). ToF cameras are well suited for small vehicles because they provide video frame rates, centimeter depth resolution, and ambient light rejection, with low computational burden and a small footprint.

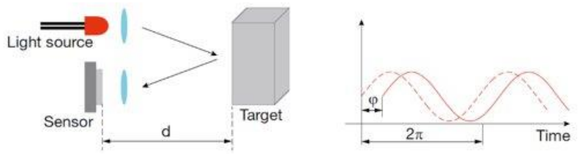


Figure 1. Indirect ToF measurement using phase of intensity modulation from Osram, "ToF Measurement using Pulse Lasers"

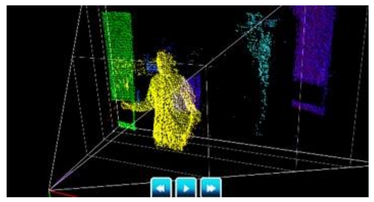


Figure 2. A sample ToF image taken with a pmdTec Flexx



System Testing

We performed a pool test of the ToF system deployed on the ROV in order to establish a clear water benchmark and confirm depth accuracy (Figure 6). In this experiment, a diver was positioned at varying distances away from the stationary camera.

A field test was also performed in which we attempted to image a Secchi disk in a highly turbid lake to assess the effects of backscatter (Figures 7 and 8). In this case, the return from the object and the return from the backscatter sum at each pixel. When the object return is strong, as in the image of the disk close to the camera, the object return dominates and the Secchi disk pattern is visible (albeit blurred). When the object return is weaker, the backscatter return dominates and the Secchi disk pattern is not clear.



Figure 8. Deploying the camera on the ROV for a lake test

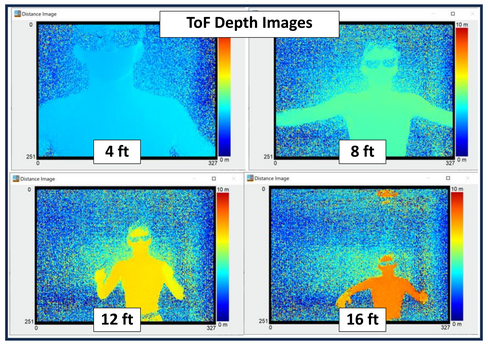


Figure 6. Clear water images from testing in pool. *Top:* Establishing shot of swimmer with BlueROV's RGB camera. *Bottom:* Depth images taken with ToF camera at various distances. The swimmer's color changes with distance.

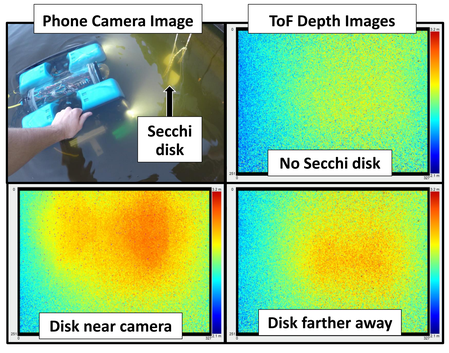


Figure 7. Scatter-limited imaging in turbid water. The Secchi disk is heavily blurred even when near the camera. As the disk moves away from the camera the backscatter dominates.

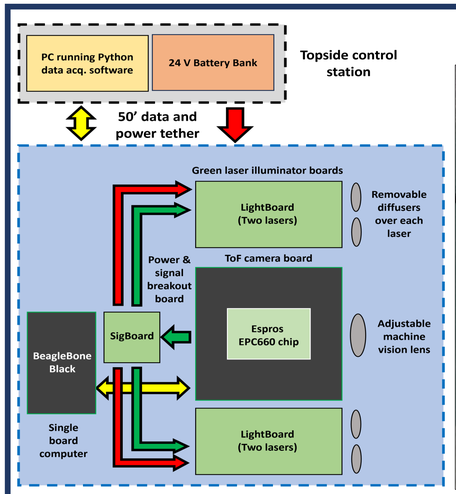


Figure 3. ROV-mounted time-of-flight system block diagram. Connection arrows are color coded by signal type: yellow for digital data, red for power flow, and green for laser timing.

Camera System Design

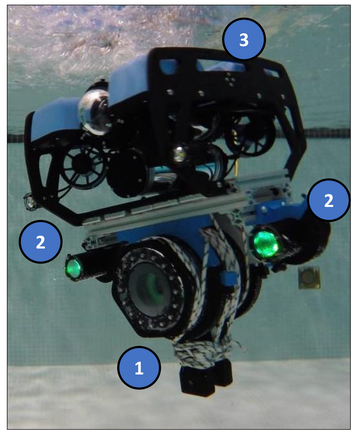
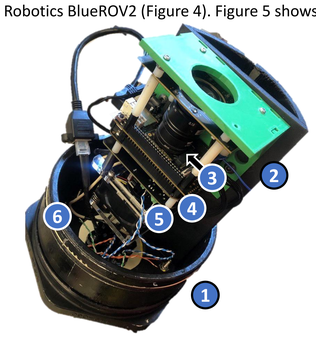


Figure 4. *ToF system mounted on BlueROV deployed in a pool. Numbered items:*
1: ToF camera enclosure
2: Laser illuminator modules
3: BlueROV

Our underwater 3D camera is built around an Espros EPC660 ToF kit (Figure 3). The kit uses infrared (IR) lasers for imaging, but we modified the kit to use green lasers because IR is quickly absorbed by water. We developed two custom circuit boards: the "SigBoard" to route the camera's laser drive signal and two "LightBoards" to drive the lasers. Each LightBoard is loaded with two 525 nm green lasers. Power and control of the camera are provided by a wired tether from two 12V batteries and a PC running a custom Python data collection program.

The camera and two illuminator modules are enclosed and deployed on a Blue Robotics BlueROV2 (Figure 4). Figure 5 shows the camera stack.



1. 6" PVC housing
2. 3D printed camera sled
3. ToF camera and optics
4. Single-board computer
5. Signal breakout board
6. Fan and tether connectors

Figure 5. *The camera enclosure hardware stack*

Backscatter Removal

We performed a laboratory water tank experiment with the ToF camera in which we attempted to remove the backscatter using a phasor subtraction method. Figure 9 shows the experimental setup and the clear water baseline images. Figure 10 shows the improvement in ToF depth measurement accuracy using backscatter removal for an object 4 cz from the camera. Figure 11 shows a more extreme example in which the object is 6.5 cz from the camera. Here, correction allows the target to be seen "through" the backscatter in the ToF amplitude measurement. The uncorrected ToF depth measurement has an error of over 5 cz, but this is reduced to 0.3 cz after correction. These promising results suggest backscatter removal should improve both object detection and ranging.

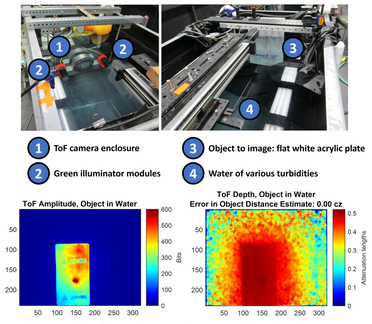


Figure 9. Experimental setup for backscatter subtraction testing, and clear water baseline.

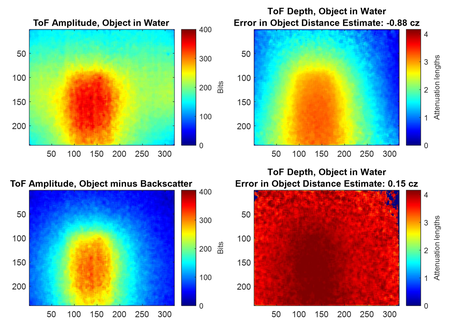


Figure 10. Backscatter removal results when the object is 4 cz from the camera.

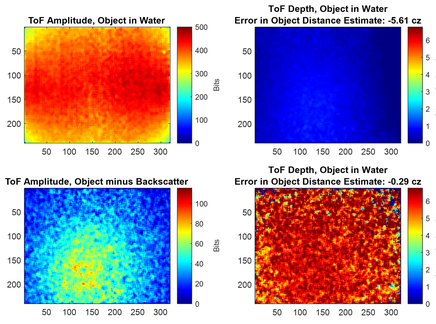


Figure 11. Backscatter removal results when the object is 6.5 cz from the camera.

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